

EXPLAINABLE AND BIAS-AWARE MULTI-LINGUAL SENTIMENT ANALYSIS WEB PLATFORM FOR INDIAN CODE-SWITCHED SOCIAL MEDIA TEXT

A mini-project synopsis report

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in partial fulfillment of the requirement for the award of the degree of

MASTER OF DATA SCIENCE



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JUNE 2026

1. Introduction

India has over 600 million internet users who naturally mix regional languages with English within the same sentence, producing Hinglish (Hindi-English), Tanglish (Tamil-English), Manglish (Malayalam-English), and Tenglish (Telugu-English). These code-switched varieties dominate Indian social media yet every existing sentiment tool including Google Cloud NLP, AWS Comprehend, and VADER is built exclusively for monolingual text and fails on them entirely. LinguaLens is a full-stack web application that accurately classifies sentiment and emotion across all four Indian code-switched language pairs, integrates SHAP token-level explainability, and provides a bias audit panel — making it the first accessible, explainable, and fairness-aware sentiment platform for Indian code-switched text.

2. Need / Motivation

Existing tools fail Indian code-switched text in three specific ways: (1) Accuracy failure — emotionally loaded words like bakwaas, romba, adipoli, and baagundi are ignored as unknown tokens; (2) Transparency failure — predictions carry no explanation of which words drove the result; (3) Fairness failure — no tool checks whether predictions are biased against Hindi-dominant versus Tamil-dominant text. This creates a systemic data failure affecting every brand, researcher, and healthcare organisation that depends on Indian social media data.

3. Problem Statement

- 700 million+ Indians express opinions daily in code-switched language that no existing AI tool correctly reads.
- Brands, researchers, and healthcare workers make decisions on fundamentally incorrect Indian sentiment data.
- No web platform exists combining Indian code-switched sentiment analysis with explainability and fairness auditing.

4. Domain / Application Area

Domain: Social Media Analytics and Digital Marketing Intelligence.

Why selected: Social media is where code-switching is most prevalent and commercially significant for Indian audiences. The domain directly maps to all four course outcomes — web

development (CO1), data science integration (CO2), problem solving (CO3), and data ethics (CO4).

Who benefits: E-commerce platforms, film industries (Kollywood, Tollywood, Mollywood), healthcare organisations, political researchers, and 700 million Indian social media users whose natural language is finally understood by AI.

5. Business Problems

1. Misread Customer Reviews: E-commerce platforms misclassify Hinglish/Tanglish product reviews, leading to incorrect product quality signals and poor business decisions.
2. Invisible Brand Sentiment: Indian social media campaigns generate code-switched feedback that analytics tools cannot process, leaving brands blind to real audience perception.
3. Missed Mental Health Signals: Healthcare organisations miss crisis expressions in code-switched Indian text, reducing effectiveness of early intervention systems.
4. Unread Regional Audience Reactions: Film studios cannot measure genuine sentiment from Tanglish, Tenglish, and Manglish social media posts about their content.
5. Biased AI Predictions: No sentiment system audits equitable performance across language-dominant text groups, causing silent discrimination against certain Indian language speakers.
6. Inaccessible Research Models: Academic code-switched NLP models exist only as code repositories with no usable interface for non-technical practitioners.

6. Proposed Methodology

Data Collection: SemEval-2020 Task 9 (14,000 Hinglish tweets), TamilMixSentiment (15,744 Tanglish comments), DravidianCodeMix (20,000+ Manglish/Tenglish comments) from GitHub, HuggingFace, and Kaggle.

Preprocessing: Emoji-to-text conversion, URL/mention removal, Unicode block detection for Devanagari/Tamil/Malayalam/Telugu mixed with Roman script, language tagging, and label harmonisation to Positive/Negative/Neutral.

Algorithms: Transfer learning with XLM-RoBERTa (3-class sentiment); distilroberta-base (5-class emotion); SHAP for token-level explainability; differential F1 for bias auditing; confidence thresholding at 60%.

Web Development: Flask REST API (POST /api/analyze, POST /api/batch, GET /api/history); Bootstrap 5 + Chart.js frontend; SQLite database; Google Colab T4 GPU with ngrok tunnelling.

Testing: 30 code-switched sentences for unit testing; Postman for API endpoint testing; 100-row CSV for batch testing; response time under 3 seconds.

7. Tools and Technologies

Category	Tool / Technology
Programming Language	Python 3.10
ML / DL Framework	PyTorch, HuggingFace Transformers
Models	XLM-RoBERTa, distilroberta-base
Explainability	SHAP (SHapley Additive exPlanations)
Backend	Flask 3.x, Flask-CORS
Frontend	HTML5, Bootstrap 5, Chart.js, Vanilla JS
Database	SQLite3
Data Processing	Pandas, NumPy, langdetect, emoji
Environment	Google Colab (T4 GPU), pyngrok, Git/GitHub

8. Product Delivery Mode

Web Application — accessible via any modern browser through a Flask server with ngrok tunnelling, delivering all features through a responsive Bootstrap 5 interface.

9. Expected Features

- **Single-Text Analyser:** Real-time sentiment, emotion, confidence score, and language tag for any code-switched input.
- **SHAP Explainability:** Colour-coded word-level highlights showing which tokens drove each prediction.
- **Batch CSV Processing:** Upload and analyse up to 1000 rows with downloadable results and summary statistics.
- **Bias Audit Dashboard:** Differential F1 scores across language-dominant segments with fairness gap flagging.

- Analytics Dashboard: Sentiment distribution, emotion breakdown, and language mix Chart.js visualisations.
- Query History: Paginated table with search, filter, and row-level deletion for data minimisation.
- Confidence Warning: Uncertainty badge for predictions below 60% confidence threshold.

10. Proposed Architecture / Workflow

Layer	Component	Function
Input	Web UI / REST API	Accepts text or CSV from browser or API call
Preprocessing	Pipeline Module	Language detection, script handling, tokenisation
Model	XLM-RoBERTa	3-class sentiment + 5-class emotion + confidence score
Explainability	SHAP Explainer	Token-level attribution scores mapped to input words
Output	Flask + SQLite	JSON response, dashboard charts, history, bias audit

11. Expected Outcome

- Fully functional web application analysing sentiment and emotion in all four Indian code-switched language pairs.
- REST API with endpoints for single-text analysis, batch CSV processing, and paginated history.
- SHAP explainability UI with colour-coded word-level attribution for every prediction.
- Bias audit dashboard flagging performance gaps across language-dominant text segments.
- Documented open-source GitHub repository with complete source code and demo video.

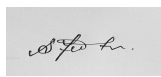
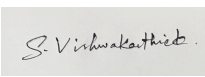
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Student Declaration

We hereby declare that the proposed project work is original and will be carried out under the guidance of the faculty mentor.

Student Signature(s):

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Date: 01- 07- 2026

Faculty Guide Signature:

